Mathematical Foundations of Universal Admissibility

Coding the Macroscopic Invariant Structure

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Abstract This report develops a registry-based framework for distinguishing local admissibility certificates from global realizability conditions across discrete computation, graph topology, feedback dynamics, delay systems, and quantum operations. In each domain, a local test—such as a continuation quotient, a declared 2-cell relation, a spectral loop-closure condition, or Gram-matrix positivity—defines a constrained boundary of allowable behavior. The analysis demonstrates that these local certificates do not generally determine the macroscopic object of interest: residual homology requires a completed relation complex; a local instability threshold does not determine the existence or basin of a stable periodic orbit; and a deterministic quantum map may become feasible only after the operational class is expanded to a heralded trace-nonincreasing process. The missing completion is formalized in each case, and an "admissibility registry" is proposed to record the state space, equivalence relation, local certificate, required global object, and resource coordinates. The framework is intended as a reproducible software specification: every registry entry is paired with declared assumptions, executable tests, and explicit falsifiers.

1. Introduction: The Local-to-Global Separation Principle

The transition from localized computational thresholds to the global topological organization of physical and informational systems requires a precise mathematical framework devoid of ad hoc modeling. The underlying algorithmic structure of reality—whether analyzed through the discrete lens of finite state automata, the continuous manifolds of delay differential equations, the geometric bounds of quantum information, or the topological constraints of graph configuration spaces—can be understood not as a maximal collection of theoretical distinctions, but as a minimum stable distinction structure. This structure is what remains after all admissible mathematical identifications and boundary conditions are rigorously imposed.

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However, a fundamental category error persists in applied mathematics, theoretical computer science, and physics: treating a local mathematical derivative or threshold as identical to the global geometric object it permits. The central mathematical principle established in this report is formalized as follows:

A local threshold can certify a local transition; it cannot, by itself, certify the global object that makes that transition observab

To code to the fundamental limits of computation and dynamics, one must evaluate an exact mathematical registry that classifies these boundaries. This requires stating the allowed class of transformations, calculating the local boundary, identifying what is strictly not represented by that local calculation, and systematically constructing the global object required for the actual physical or computational claim. This report formalizes the missing completion across four sharply bounded domains, providing a unified formal core and a strictly auditable registry.

2. Formal Core

To align several distinct domains of "admissibility" under one design discipline, a common mathematical object is defined to model the state, transformation, and resource limits of any given system. We define the universal admissibility object as the tuple:

$$\mathfrak{A} = (\mathcal{S}, \mathcal{T}, \sim, \mathcal{R}, \mathcal{O})$$

where:

- $\mathcal{S}$: state or description space (the total space of available configurations).
- $\mathcal{T}$: admissible transformations (the allowed dynamic or logical operators).
- $\sim$: equivalence or quotient relation (the topological or functional identifications).
- $\mathcal{R}$: retained resource coordinates (the required spatial dimensions, physical delays, or ancilla environmental parameters).
- $\mathcal{O}$: observables or output contract (the macroscopic structure that is mathematically guaranteed).

Within this universal framework, the underlying principle dictates that:

$$\text{Local admissibility} \neq \text{global realizability}$$

A local certificate evaluates a constrained boundary of allowable behavior. Such a certificate might take the form of a Jacobian crossing the imaginary axis, a positive-semidefinite matrix difference, a commuting logical square, or a localized quotient closure. Conversely, a global certificate is the complete mathematical manifestation that renders the phenomenon usable, such as an attracting invariant set in phase space, an explicitly generated homology class, a physically completed quantum instrument, or a finite minimal automaton quotient across all operational gaps. The transition from local certification to global realization defines the strict burden of proof required for establishing macroscopic existence.

3. Four Sharply Bounded Case Studies

3.1 Case Study 1: Graph Configuration Complex and Homological Defects

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When a discrete computational domain operates, it functions by systematically identifying equivalent states to prevent representational redundancy from masquerading as actual physical divergence. The formalism for establishing these fundamental identifications relies heavily on the homology of graph configuration spaces and their corresponding topological invariants. Configuration spaces of points are pivotal in modeling collision-free motion planning for autonomous agents, a problem domain deeply connected to geometric group theory and the topological complexity of systems.

3.1.1 Exact Graph Construction and Betti Notation

In finite token-configuration complexes, a state graph possesses a 1-skeleton representing distinct local transitional moves. The relation layer of this space is formed by formally declaring 2-cell predicates. Consider the arithmetic group:

$$G_n = \mathbb{Z}/2^n\mathbb{Z}$$

with the generating set:

$$\{2^0, \dots, 2^{n-1}\}$$

For any index $i < n-1$, the generator provides a directed edge:

$$x \mapsto x + 2^i$$

For the top generator $i = n-1$, the operation:

$$x + 2^{n-1} \equiv x - 2^{n-1} \pmod{2^n}$$

establishes an undirected, self-inverse edge, yielding a simple topological graph.

To maintain strict notational consistency, the first Betti number is denoted as $b_1(X)$, reserving β_1 solely for contexts where it is explicitly redefined as the first Betti number in specialized topological configurations. For a connected finite graph, the exact algebraic calculation of the Betti number is:

$$b_1(X) = |E| - |V| + 1$$

Under the specific edge-count convention where each generator edge is counted exactly once after identifying the self-inverse top-bit edges, the vertex and edge counts evaluate to:

$$|V| = 2^n, \qquad |E| = (n-1)2^n + 2^{n-1}$$

Therefore, the baseline Betti number calculates to:

$$b_1(X) = (n-2)2^n + 2^{n-1} + 1$$

This algebra is rigorously reproducible solely if the graph is connected and the top-generator symmetry is explicitly mapped. The first Betti number fundamentally quantifies the number of independent cycles in the graph, forming a baseline for the higher-order relations and the broader study of graph braid groups.

3.1.2 State-Dependent Commuting Predicate

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To constrain the state space, 2-cell predicates are declared over commuting operations. When two operations commute, they are bounded by a 2-cell, effectively closing the loop. For a pair of generator indices $i < j$ (representing ordinary integer bit positions) and a local vertex x, the square $r_{ij}(x)$ features vertices $x, x + 2^i, x + 2^j$, and $x + 2^i + 2^j$. This square is declared "disjoint" exactly when the two operations modify distinct bits at vertex x.

The disjointness constraint is strictly state-dependent and can be verified via the boolean logic condition (the XOR/AND test), which functions as the exact bit-vector encoding of the mathematical support-set test:

$$(x \oplus (x + 2^i)) \wedge (x \oplus (x + 2^j)) = 0 \quad \Leftrightarrow \quad \mathrm{supp}_i(x) \cap \mathrm{supp}_j(x) = \varnothing$$

where the support set is defined as:

$$\mathrm{supp}_i(x) = \{q\text{:bit } q \text{ changes under } x \mapsto x + 2^i \pmod{2^n}\}$$

Operationally, this means the addition $x + 2^i$ in $\mathbb{Z}/2^n\mathbb{Z}$ must not induce a carry chain that propagates into bit j. This holds if and only if there exists a zero bit between i and $j - 1$; mathematically, some intermediate bit index $q \in \{i, \dots, j - 1\}$ must satisfy $x_q = 0$.

To assure consistency across all configurations and evaluations, the following convention decisions are explicitly established:

- **All additions:** reduced modulo 2^n.
- **All bit positions:** represented by the canonical n-bit word in $\{0,1\}^n$.
- **All support comparisons:** use those canonical n-bit representatives.
- **Top-bit generator:** undirected as a graph edge, but evaluated by the same canonical modular-addition support map.

Wraparound is governed by reduction modulo 2^n; bit positions remain indexed by $0, \dots, n - 1$. When $j = i + 1$, the intermediate set is merely $\{i\}$. The condition violently collapses to the requirement that $x_i = 0$. If $x_i = 1$, adding 2^i flips bit i and carries directly into j, violating disjointness.

3.1.3 The Homological Defect and Discrete Morse Theory

The residual topological dimension—representing the unclosed, macroscopic state distinctions—is measured by the algebraic quotient of these boundary cycles. Defining Z_1 as the cycle space of the shared 1-skeleton, the boundary subspaces are formalized as follows:

- $B_D \subseteq B_A \subseteq Z_1$
- B_D is the boundary span of the declared 2-cells.
- B_A is the boundary span of all mathematically possible squares.

The structure operates under the strict inclusions mapping the localized logical checks to the true geometrical boundaries. The homology of the declared and all-square complexes evaluates to:

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$$H_1(D) = Z_1/B_D, \qquad H_1(A) = Z_1/B_A$$

From these definitions, the homological defect is derived as the descent identity:

$$\dim H_1(D) - \dim H_1(A) = \dim(B_A/B_D)$$

This quotient is strictly a homological defect based on boundary subspaces. Determining this defect requires translating localized combinatorial conditions into a global topological evaluation. This is where discrete Morse theory, particularly the Farley-Sabalka discrete gradient field, becomes computationally indispensable. In this algorithmic approach, the cells of the complex are sorted into a Hasse diagram, and a discrete gradient field defines non-overlapping pairs of collapsible and redundant cells. Cells that cannot be paired in this manner are deemed*critical*.

According to discrete Morse theory, the homology of the underlying space is completely captured by a simplified Morse chain complex constructed strictly from these critical cells. By identifying exactly where $b_1(D)$ diverges from local expectations—meaning an abundance of critical 1-cells that are not bounded by critical 2-cells—the defect geometrically isolates precisely what physical or logical structure is not represented by a localized calculation.

Co Reproducibility Block: Graph Configuration Complex

- **MODEL ID:** $\mathbb{Z}/2^n\mathbb{Z}$ discrete graph configuration with self-inverse top-bit generator.
- **PARAMETERS:** Total bit width n. Operations modulo 2^n. Edge counting assumes undirected top-bit consolidation.
- **INITIAL CONDITIONS:** Base 1-skeleton generating Z_1 with declared 2-cell boundary B_D.
- **SOLVER:** Exhaustive boundary matrix nullspace enumeration via Farley-Sabalka discrete Morse algebraic reduction.
- **OBSERVABLE:** Computed $H_1(D)$ Betti number and declared-vs-all-square defect $\dim(B_A/B_D)$.
- **OUTPUT:** Exact raw integer nullspace ranks.
- **FALSIFIER:** Identification of any vertex x where $j = i + 1$, $x_i = 1$, and the declared logical predicate evaluates to 0 without triggering a carry cascade collision in the boundary matrix.

3.2 Case Study 2: Carry / Reader-Cut Quotient in Streaming Transducers

To map discrete local structural rules into macroscopic calculation, the functional state space of an automaton must be derived as the rigorous mathematical quotient of descriptions necessitating identical future continuations.

3.2.1 Streaming Transducers and Myhill-Nerode Continuations

In the classical theory of formal languages, the Myhill-Nerode theorem proves that a formal language is regular if and only if its associated right-invariant equivalence relation divides all possible strings into a finite number of classes. This right congruence dictates that two prefixes are equivalent if and only if appending any arbitrary suffix yields identical membership outcomes.

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However, when extending this logic to generalized arithmetic streaming transducers that evaluate multi-operand addition over an arbitrary input sequence, classical deterministic finite-state limits are rapidly exceeded. Streaming String Transducers (SSTs) manipulate registers using copyless updates to store partial outputs, necessitating state minimization over both control states and memory registers. Because the transducer writes to an output tape while maintaining a finite set of variables updated in an additive or concatenative manner, its local transformation structure differs fundamentally from simple accept/reject automata.

For a reader cut tracking modular addition:

$$(a, b) \mapsto a + b \pmod{2^n}$$

over an input sequence of positions S, two partial reads are equivalent exactly when every conceivable future completion yields identical, not-yet-emitted outputs. This equivalence defines the minimal Myhill-Nerode continuation quotient class Φ_S for rational functions. Because the local operator updates the state entirely independently of any underlying unread variable outside the current maximal read runs, the reader's minimal continuation quotient is strictly closed under its own local evolution. A transducer is algebraically minimal when any equivalent realization can be mapped as a quotient of a subobject, reflecting the minimal register complexity required to evaluate independent arithmetic operations across gaps.

3.2.2 Capacity Factorization Threshold

The structural limitation of this congruence is quantified by evaluating the threshold at which the macroscopic capacity of a state machine factorizes over its unread gaps. The computational bound evaluates the exact parameter range separating entangled arithmetic debts from mathematically independent gap capacities.

In the enumerated parameter range and under the declared streamed-output convention, every tested cut satisfying:

$$b^g \geq m$$

exhibited exact factorization:

$$F(S) = \prod_R f\ (R)$$

where b is the numerical base, m is the fixed operand count, and g is the unread gap length in digit positions. When the condition $b^g < m$ occurs, the amount of data required to track overlapping carry obligations exceeds the capacity of the unread segment. No single sequence of gap words can realize a constant mathematical transfer, causing the mathematical carry debts to overlap across gaps. Under these conditions, the product law $\prod_R f\ (R)$ wildly overcounts the true combinatorial capacity $F(S)$, forcing a unified macroscopic state description. This separability threshold evaluates independently of the total word width n, subject to standard bounded streaming conventions.

Co Reproducibility Block: Carry / Reader-Cut Quotient

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- **MODEL ID:** m-operand base-b arithmetic streaming transducer evaluated over arbitrary reader cuts.
- **PARAMETERS:** Base b, operand count m, unread gap length in digits g, total word width n.
- **ENUMERATION DOMAIN:$b \in \{2, 4\}n \in \{8, 16, 32\}$$**g \in \{1, 2, 3, 4\}$ cut-family definition: regular alternating read/unread segmentation sequences.
- **INITIAL CONDITIONS:** Empty read set $S_0 = \emptyset$. Outputs are strictly streamed during evaluation, not post-held.
- **SOLVER:** Exact finite algebraic enumeration of generalized equivalence classes mapping identical future emitted outputs.
- **OBSERVABLE:** Raw global state count $F(S)$ compared to the multiplicative gap quotient $\prod_R f\ (R)$.
- **OUTPUT:** Exact integer capacities and binary match arrays.
- **FALSIFIER:** A specific read cut configuration satisfying the separability threshold $b^g \geq m$ for fixed b, m, g where the global quotient fails to factorize ($F(S) \neq \prod_R f\ (R)$) under exact enumeration.

3.3 Case Study 3: Feedback/DDE Local-to-Global Gap

Continuous physical networks governed by temporal delays and feedback loops perfectly illustrate the danger of confusing local derivatives with macroscopic geometric realizability. The local invariant root governs feedback, phase closure, and wave transport by calculating exact linear instability boundaries, but remains blind to macroscopic continuations, hysteresis, and basin boundaries.

3.3.1 Linear Onset and the Spectral Registry

For a linear onset candidate, the mathematical mandate for phase closure dictates that any signal returned through a cyclic graph must arrive with a target phase Φ and a round-trip magnitude of unity. The phase and magnitude bill calculates the critical external loop gain G_{crit} and operating frequency ω via the transfer descriptor equations:

$$\sum_k \arg H_k(i\omega) + \omega\tau = \Phi, \qquad G\prod_k |\,H_k(i\omega)| = 1$$

To place highly distinct physical phenomena into one auditable registry, exact loop architectures, normalizations, and bifurcation classes must be aggressively qualified:

- **Zero-frequency feedback marginality:** A declared two-stage normalized feedback loop may satisfy a zero-frequency marginal loop condition at $G = 1$. This is distinct from finite-frequency Hopf onset. Whether the associated nonlinear model undergoes a saddle-node, pitchfork, transcritical bifurcation, latching transition, or no codimension-one bifurcation at all depends on its equilibrium equation and nonlinear terms; the linear zero-frequency condition alone does not decide that classification.

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- **Gain Parameterization:** For 3 identical first-order lag stages where $L(s) = \frac{G}{(1+s/\gamma)^3}$, if the target phase is π, a normalized loop condition yields a global external scalar loop gain:

$$G_{\text{crit}} = \sec^3\left(\frac{\pi}{3}\right) = 8$$

Conversely, if the gain is parameterized strictly as a per-stage amplifier G_{stage}^3, the threshold is exactly:

$$G_{\text{stage,crit}} = \sec\left(\frac{\pi}{3}\right) = 2$$

Both are correct, but the parameterization convention must be locked prior to evaluation.

- **Resonator Magnitude Claim:** Declared resonator architecture. Let the loop transfer function be:

$$L(s) = G\,\frac{1}{1+s/\omega_n}\,\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

with the external scalar G placed outside the cascaded lag–resonator factors, target phase branch $-\pi$, and:

$$Q = \frac{1}{2\zeta}$$

Under this exact normalization, solving the phase and magnitude equations yields the stated G_{crit} formula:

$$G_{\text{crit}} = 4\zeta(1+\zeta) = \frac{2}{Q} + \frac{1}{Q^2}$$

The resonant factor contributes phase rotation and, near its resonance regime, can reduce the external scalar gain required by the declared loop-closure equation. The extent and sign of that reduction depend on the exact transfer normalization and loop architecture.

3.3.2 Delay Differential Equations and the Attractor Atlas

While the spectral equations perfectly locate a linear onset candidate where a characteristic root reaches the imaginary axis, they are mathematically blind to the resultant global structure. Delay Differential Equations (DDEs) model phenomena whose derivatives depend on historical phase states, expanding the functional space to the infinite-dimensional history space:

$$C^0([-\tau, 0]; \mathbb{R}^n)$$

Unlike Ordinary Differential Equations (ODEs) where the state is merely a finite-dimensional vector, the integration of a DDE requires a functional initial condition representing a history segment. The characteristic transcendental equation identifies a pair of complex roots $\pm i\omega_0$ crossing the imaginary axis. The transition to a finite-dimensional Center Manifold enables the derivation of the normal form equation and the first Lyapunov coefficient, l_1. The sign of l_1 dictates the criticality of the local Hopf bifurcation:

- If $l_1 < 0$, the bifurcation is supercritical, birthing a continuous, stable periodic orbit.
- If $l_1 > 0$, the bifurcation is subcritical, producing an unstable, repelling limit cycle.

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In the subcritical regime, crossing the local instability threshold ejects the trajectory entirely from the local neighborhood. The isolated Jacobian derivative and the calculation of l_1 at the origin cannot predict where the trajectory lands, nor the amplitude of the emerging remote orbit, nor the hysteresis interval. The existence of a subcritical Hopf absolutely mandates that a macroscopic structure exists further out in the global phase space to capture the trajectory, but the local certificate cannot name the global event in advance.

Thus, a scalar critical parameter is merely a local coordinate. The necessary completion is the Attractor Atlas, defined computationally as the global geometrical object:

$$\mathcal{A}(\mu) = \{\text{equilibria,periodic orbits,stability,basins,boundary connections}\}$$

Possible remote events include: saddle-node of periodic orbits, homoclinic connection, heteroclinic connection, collision with another invariant set, or unbounded continuation. Global periodic-orbit continuation, executed via solvers such as DDE-BIFTOOL, is strictly required for statements regarding bistability, folds of cycles, and basin boundaries. DDE-BIFTOOL leverages collocation methods—often utilizing piecewise polynomial representations and Chebyshev nodes—to rigorously approximate the infinite-dimensional system and track the stability of limit cycles through subsequent bifurcations.

Co Reproducibility Block: Feedback/DDE Local-to-Global Gap

- **MODEL ID:** Single scalar Delay Differential Equation with linear delayed feedback and cubic saturation nonlinearity.
- **PARAMETERS:** Sweep of delay parameter $\tau \in [0.5,5.1]$; explicit constant normalized gain.
- **INITIAL CONDITIONS:** 12 uniformly sampled constant history functions evaluated across the infinite-dimensional interval $C^0([-\tau, 0]; \mathbb{R}^n)$.
- **SOLVER:** DDE-BIFTOOL v3.1.1 (MATLAB), adaptive step control with Chebyshev node collocation for center manifold reduction and numerical continuation.
- **OBSERVABLE:** Diagnosis of stable limit cycle attraction; calculation of first Lyapunov coefficient l_1.
- **OUTPUT:** Raw numerical phase trajectories and fractional basin hit metrics.
- **FALSIFIER:** A theorem proving that, for the declared model class, the stated local certificate—critical eigenvalues, crossing direction, and first Lyapunov coefficient—uniquely determines the complete global attractor atlas, including the existence, stability, continuation, and basin geometry of all relevant periodic orbits.
- **SUPPORTING COUNTEREXAMPLE PROTOCOL:** Construct two systems with identical declared local Hopf data but distinct global periodic-orbit continuation, remote bifurcations, or basin geometry. Such a pair supports the local-to-global separation principle. (Empirical note: Under the declared 12 sampled constant history functions and stated numerical integration settings, no sampled trajectory converged to the identified periodic orbit below the tested range near $\tau \approx 1.5$. This supports basin dependence within the tested protocol; it does not prove nonexistence of the periodic orbit or zero basin measure below that parameter value.)

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3.4 Case Study 4: Two-State Quantum Cloning and Gram-Matrix Positivity

Quantum informational operations provide a stark demonstration of how expanding an operational class is structurally required to complete an otherwise strictly impossible local action. Any completely positive trace-preserving (CPTP) map must rigidly adhere to Gram matrix inequalities regarding the geometric overlap of quantum states.

3.4.1 Deterministic Impossibility and Matrix Eigenvalues

For two pure states with overlap $x = \langle\psi_1|\psi_2\rangle \in [0,1]$, deterministic exact cloning attempts to map the source states to the separable target states $|\psi_1\psi_1\rangle$ and $|\psi_2\psi_2\rangle$. The initial source Gram matrix is:

$$X^{(1)} = \begin{pmatrix} 1 & x \\ x & 1 \end{pmatrix}$$

The cloned target Gram matrix evaluates as:

$$X^{(2)} = \begin{pmatrix} 1 & x^2 \\ x^2 & 1 \end{pmatrix}$$

Deterministic cloning requires the relevant Gram-matrix difference to be positive semidefinite (PSD), satisfying the fundamental constraint that distinguishability cannot increase under a trace-preserving quantum operation. At a fixed success probability of $p = 1$, the difference is:

$$X^{(1)} - X^{(2)} = \begin{pmatrix} 0 & x - x^2 \\ x - x^2 & 0 \end{pmatrix}$$

The eigenvalues of this matrix are analytically determined as:

$$\lambda_{\pm} = \pm(x - x^2)$$

For any non-orthogonal, non-identical pair of states where $0 < x < 1$, the scalar value $x - x^2$ is inherently positive. Consequently, the minimum eigenvalue is strictly negative:

$$\lambda_{\min} = -(x - x^2) < 0 \quad (0 < x < 1)$$

Because the minimum eigenvalue is negative, the matrix fundamentally fails the positive semidefinite condition. This exact algebraic failure is the local obstruction that mathematically emits the deterministic no-cloning theorem.

State Overlap x	Scalar (x–x2)	Minimum Eigenvalue at p=1	Deterministic Cloning
0.0 (Orthogonal)	0.00	0.00	Admissible
0.2	0.16	-0.16	Not Admissible
0.5	0.25	-0.25	Not Admissible

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State Overlap x	Scalar (x−x2)	Minimum Eigenvalue at p=1	Deterministic Cloning
0.8	0.16	-0.16	Not Admissible
1.0 (Identical)	0.00	0.00	Admissible

3.4.2 Probabilistic Limits and Ancillary Dimension Constraints

For the stated task—exact deterministic $1 \to 2$ cloning of two distinct nonorthogonal pure states—the impossibility holds under the CPTP operational class. However, changing the operational class from a deterministic CPTP map to a heralded/probabilistic completely positive trace-nonincreasing map (CP-TNI) alters the feasible boundary. A successful branch can be executed provided it is paired with a failure branch inside a mathematically completed instrument:

$$\mathcal{E}_{\text{succ}} + \mathcal{E}_{\text{fail}} = \mathcal{E}_{\text{CPTP}}$$

By treating the success probability p as a free variable to optimize, exact equal-success-probability cloning establishes the known probabilistic optimum:

$$p_{\max} = \frac{1}{1+x}$$

This structural adjustment confirms the class extension:

$$\text{CPTP} \subsetneq \text{CP-TNI}$$

Additionally, ensuring complete positivity demands physical resources. The Stinespring Dilation Theorem dictates that any noisy, irreversible process on a quantum system can be understood as a perfectly reversible unitary evolution on a larger system comprising the principal Hilbert space tensor-producted with an ancilla environment. The required environment dimension d_a acts as a structural resource boundary.

Through the Choi-Jamiołkowski isomorphism, a quantum channel is mapped to a positive semidefinite bipartite state known as the Choi state. The environment dimension boundary mathematically saturates exactly when d_a reaches the rank of this Choi matrix:

$$d_a = \text{rank}(J)$$

Any ancilla dimension appended beyond the Choi-rank saturation operates purely as an inert resource. The local algebraic certificate (positive semidefiniteness) inherently demands this global physical instrument—the Stinespring dilation—to resolve the operational claim. Without explicitly tracking the necessary dimension expansion, the model remains purely abstract.

3.4.3 Separating Finite Sampling from Thermodynamic Erasure

In establishing physical resource constraints, one must explicitly separate thermodynamic arguments from finite-counting statistical limits, as they originate from fundamentally different bounds:

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- **Finite Sampling:** The generalized fluctuation scale scaling as:

$$\Omega^{-1/2}$$

constitutes a statistical scaling statement. It strictly represents the uncertainty and estimation price bounded by relevant independence and effective-sample-size assumptions.

- **Thermodynamic Erasure:** Conversely, Landauer's bound:

$$kT\ln 2$$

evaluates the physical thermodynamic and entropy price of logically irreversible reset under specific conditions, such as coupling to a heat bath at temperature T.

Landauer's limit is not mathematically derived from generic finite-count fluctuations. They belong to distinct phenomenological classes, separating heat dissipation boundaries from statistical noise reliability.

Co Reproducibility Block: Two-State Quantum Cloning

- **MODEL ID:** Gram-matrix difference evaluation for 2-state optimal cloning.
- **PARAMETERS:** Real overlap scalar $x \in [0,1]$.
- **INITIAL CONDITIONS:** Defined pure state pairs under CPTP and CP-TNI map classifications.
- **SOLVER:** Exact linear algebra determinant and eigenvalue extraction.
- **OBSERVABLE:** Minimum eigenvalue $\lambda_{\min}$ and maximum parameter p preserving the PSD constraint.
- **OUTPUT:** Exact limits verifying $p_{\max} = 1/(1+x)$ under CP-TNI assumptions.
- **FALSIFIER:** Mathematical construction of a valid deterministic CPTP map yielding $\lambda_{\min} \geq 0$ for $x = 0.5$ at $p = 1$.

4. Registry Appendix: Admissibility Certificates and Missing Global Objects

The insights formalized in this report demand an explicit, auditable registry. The following table identifies disparate phenomena, explicitly separating their local transition boundaries from the macroscopic topological or geometric object required for global claims.

Domain	State / Equivalence Space	Local Certificate	Required Completion	Status / Assumptions
Graph Configuration	1-skeleton and declared 2-cell relations	Declared commuting squares;	Residual H_1, Betti defect,	Proved here. Requires specific edge-count

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Domain	State / Equivalence Space	Local Certificate	Required Completion	Status / Assumptions
		boolean disjointness predicate	undeclared cycle classes	convention and modulo 2^n arithmetic bounds.
Automata / Transducers	Myhill-Nerode continuation quotient	Factored partial read across unread gaps	Finite minimal quotient; global exact integer enumeration	Empirical Finding. Exact model for fixed base b, fixed operand count m, and streamed outputs.
Zero-frequency feedback marginality	Zero-frequency transfer models	Declared two-stage normalized feedback loop	$\Phi = 0, \omega = 0,$ and $(G \prod_k H_k\,(0) = 1)$	Standard Result Reproduced. Depends on nonlinear equilibrium equation terms.
Feedback Oscillation	Finite-frequency continuous phase space	Phase closure ($\Phi = \pi$) and unit loop magnitude	Existence, stability, and basin geometry of periodic attractors	Standard Result Reproduced. Per-stage gain parameterized as G_{stage}^3.
Resonator Models	Quality-factor Q parameterization	Minimal real gain G_{crit} satisfying loop closure	Resonant energy accumulation replacing active gain	Standard Result Reproduced. Assumes $Q = 1/(2\zeta)$ in 2nd-order normalized low-pass.

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Domain	State / Equivalence Space	Local Certificate	Required Completion	Status / Assumptions
Delay Equations (DDEs)	Infinite-dimensional history space $C^0([-\tau, 0]; \mathbb{R}^n)$	Characteristic roots crossing the imaginary axis; l_1	Attractor atlas $\mathcal{A}(\mu)$: periodic-orbit branches, stability, basin evidence, and remote global bifurcations	Numerical observation; requires declared continuation protocol and separately qualified basin analysis.
Quantum Operations	Isometry geometry; Gram-matrix preservation	Matrix positivity; $\lambda_{\min} \geq 0$	Completed CP-TNI instrument; Choi-rank saturation	Standard Result Reproduced. Explicitly assumes $\mathrm{CPTP} \subsetneq \mathrm{CP\text{-}TNI}$ expansion for probabilistic regimes.
Information Resources	Local state-space enumeration	Fluctuation scale $\Omega^{-1/2}$ vs $kT\ln 2$	Sampling uncertainty price vs Thermodynamic constraints	Standard Result Reproduced. Distinguishes statistical scale from irreversible logical erasure heat.

5. Conclusion

The architecture of universal admissibility dictates that a local threshold or algebraic certificate is mathematically insufficient to determine the macroscopic geometry of the phenomenon it regulates. Across graph configuration topology, continuous deterministic automata, delay differential dynamics, and quantum isometry operations, identifying where a local certificate fails highlights exactly what macroscopic object must be assembled next.

Betti dimension deviations demand completed relation complexes; Myhill-Nerode string quotients require finite algebraic closures that correctly factor overlapping arithmetic bounds; subcritical Hopf bifurcations

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absolutely mandate the computational plotting of an Attractor Atlas because the linear Jacobian contains no topological information regarding remote folds of limit cycles; and deterministic Gram-matrix negative eigenvalues dictate that local operational boundaries must expand into heralded completed instruments.

The registry does not replace domain mathematics with a universal formula. It imposes a common proof obligation across domains. For each claimed macroscopic phenomenon, the analyst must state the state space, admissible transformation class, quotient relation, local certificate, missing global object, resource coordinates, and explicit falsifier. A local threshold may establish onset, compatibility, or obstruction; it does not automatically establish a global attractor, completed relation complex, finite factorization law, or physical instrument. The admissibility registry makes that boundary executable: it records exactly what has been established, exactly what remains unproved, and the next mathematical object required to close the claim.

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